

FAIZEEN SABA NAZ

AI/ML Engineer

faizeensabakhan@gmail.com | +91 9354430271

ABOUT ME

AI/ML Engineer and M.Tech Computer Science postgraduate with hands-on experience in creating, fine-tuning, evaluating, and deploying machine learning, deep learning, and NLP models. Skilled in Python, PyTorch, Transformer/BERT architectures, transfer learning, data preprocessing, model evaluation, explainable AI, and AI-enabled robotics workflows. Interested in applying AI/ML to medical imaging, Hinglish NLP, forecasting, robotics automation, and intelligent software systems.

EDUCATION

Jamia Hamdard, New Delhi, India

M.Tech in Computer Science and Engineering | Aug 2024 - Present | CGPA: 8.19/10

Jamia Hamdard, New Delhi, India

B.Tech in Computer Science and Engineering | 2021 - 2024 | CGPA: 7.2/10

Jamia Millia Islamia, New Delhi, India

Diploma in Computer Engineering | 2017 - 2020 | 69.99%

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript
- **AI/ML & NLP:** Model creation, fine-tuning, evaluation, deployment workflows, Transformers, BERT variants, NLP preprocessing
- **Databases:** SQL
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Pandas, NumPy, PyTorch
- **Web Technologies & Tools:** HTML, CSS, Tailwind CSS, Material UI, Git
- **Software & Platforms:** MATLAB

ACADEMIC SKILLS

- Machine Learning
- Artificial Intelligence
- Natural Language Processing
- Data Structures, Database Management System, Operating Systems

EXPERIENCE

AI/ML Engineer / Robotics AI Developer

Chilly Robotics Kitchen, New Delhi, India | March 2024 - March 2026

- Created and integrated AI/ML-driven robotics modules using sensor data, preprocessing, and real-time inference workflows to improve responsiveness and decision accuracy.
- Fine-tuned and optimized machine-learning pipelines for robotic system behavior, improving prediction reliability and hardware-software coordination in dynamic environments.
- Deployed AI-assisted automation features into robotics applications, including model evaluation, debugging, and performance monitoring for stable operation.
- Implemented control logic and real-time data processing pipelines to connect robotic hardware with intelligent software systems.

PROJECTS

Final Year M.Tech Project: Explainable Alzheimer MRI Classification

Python | PyTorch | Transfer Learning | CNN | Grad-CAM | Medical Image Analysis

- Developed a multiclass Alzheimer's disease classification system from brain MRI images for NonDemented, VeryMildDemented, MildDemented, and ModerateDemented stages.
- Fine-tuned AlexNet, GoogLeNet, MobileNetV2, EfficientNet-B0, and ResNet50 using transfer learning with preprocessing, augmentation, and leakage verification.
- Validated the models on OASIS, Kaggle held-out, and third external MRI datasets to measure cross-dataset robustness and generalization.

- Achieved best results with EfficientNet-B0: 99.34% OASIS accuracy, 96.98% Kaggle held-out accuracy, and 95.60% external dataset accuracy.
- Integrated Grad-CAM and saliency maps to explain model predictions and highlight disease-relevant MRI brain regions.

Hinglish Hate Speech Detection using Transformer Model

Python | PyTorch | NumPy | Pandas | Matplotlib

- Developed a Transformer-based hate speech detection system for Hinglish social media text.
- Fine-tuned multiple BERT variants including mBERT, XLM-RoBERTa, DistilBERT, and BERT-base.
- Trained and evaluated models on approximately 76,000 labeled samples.
- Achieved approximately 98% accuracy with high F1-score using multilingual BERT.
- Implemented complete NLP pipeline including preprocessing, tokenization, fine-tuning, and evaluation.
- Evaluated performance using accuracy, precision, recall, F1-score, and ROC-AUC.

Enhancing Equity Analysis and Forecasting Using Attention-Based Models

Python | PyTorch | NumPy | Pandas | Matplotlib

- Designed and implemented Transformer-based architectures for stock price forecasting.
- Integrated OHLCV data, technical indicators, and FinBERT-based sentiment analysis from financial news.
- Achieved Directional Accuracy (DA) of 90.1% and improved risk-adjusted returns.
- Provided interpretable attention visualizations to identify influential market drivers.
- Benchmarked models against existing approaches and demonstrated improved forecasting performance.

RESEARCH PUBLICATION

Improving Diabetes Diagnosis: A Study on Machine Learning and Model Fine-Tuning

Presented at IEEE ICAIHI-2025, 2nd International Conference on Artificial Intelligence for Innovations in Healthcare Industries, organized by Shri Shankaracharya Institute of Professional Management and Technology (SSIPMT), Raipur, Chhattisgarh, India, December 2025.

Multilingual BERT-Based Hate Speech Detection in Hinglish Social Media Content Analysis

Published in International Journal of Scientific Research in Engineering and Management (IJSREM), ISSN: 2582-3930. [Paper Link](#)